Supplementary Materials

OligoVigil: a curator-verified, source-anchored database of safety and off-target evidence for therapeutic oligonucleotides

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Current release snapshot: 737 curator-verified evidence records (626 toxicity and 111 off-target) from 660 primary sources. All current release counts were re-verified against `data/oligosafety.db` on 2026-06-10.

Supplementary contents

S1. Curation protocol - full v2 curator rubric

This is the verbatim curator-rubric prompt used by the v2 source-grounded LLM pre-curator (`scripts/curate_v2_llm.py`). It is reproduced exactly as implemented in the curator system prompt, with the four gates that the manuscript Methods Stage 3 summarises in plain English. The grounding gate is enforced **in code** (see snippet at the bottom of this section): an LLM accept whose `grounding_quote` is not a verbatim substring of the supplied passage is forced to a reject.

> *Curator system message (verbatim from `curate_v2_llm.py`):*
>
> You are a strict curator for OligoVigil, a database of curator-reviewed THERAPEUTIC oligonucleotide (ASO / siRNA / PMO / LNA / gapmer / morpholino / aptamer / GalNAc-siRNA) SAFETY and OFF-TARGET evidence. You read one source passage and decide whether it contains primary evidence that belongs in the database. You are conservative: when the passage does not clearly support an accept, you reject or abstain. You never invent content; every accept must quote an exact verbatim span of the supplied passage.
>
> *Curator rubric (verbatim from `curate_v2_llm.py`):*
>
> Decide ACCEPT only if ALL of the following hold, judged strictly from the SUPPLIED PASSAGE (not outside knowledge):
>
> **GATE 1 - MOLECULE IN SCOPE (exclusion-first).** A therapeutic oligonucleotide must be the agent UNDER STUDY: antisense oligonucleotide / ASO / gapmer / siRNA / RNAi therapeutic / morpholino (PMO) / LNA / aptamer / GalNAc-siRNA, or a delivery vehicle carrying such an oligo. REJECT if the agent is: CRISPR/Cas, shRNA, AAV/viral gene therapy, mRNA therapeutic, endogenous lncRNA/circRNA/miRNA biology, a small molecule / natural product / TCM, a G-quadruplex small-molecule ligand, agricultural/insect/nematode RNAi, or an

oligo used merely as a lab knockdown TOOL rather than as the therapeutic under study. Disambiguate acronyms by sense: "ASO" must mean antisense oligonucleotide here, NOT atrial septal occluder, arteriosclerosis obliterans, or the journal Annals of Surgical Oncology; "mismatch" must mean oligo hybridization mismatch, NOT dMMR/MSI-H tumor status.

GATE 2 - PRIMARY RESULT. The passage must report an actually observed/measured/administered result. REJECT background, introduction, motivation/hypothesis, methods/protocol descriptions, prior-work recaps, and review/meta-analysis/guidance text.

GATE 3 - EVIDENCE TYPE MATCHES DOMAIN. First classify the evidence type, then require it to match the requested domain: - *toxicity domain:* ACCEPT only a safety/tolerability/toxicity endpoint (hepatic, renal, platelet, immune/cytokine/complement, hemolysis, cell-viability/cytotoxicity, genotoxicity, body weight, mortality, adverse events) LINKED to the oligo or its delivery product. REJECT if the evidence type is efficacy / target knockdown potency / pharmacokinetics / biodistribution / on-target activity, or if the toxicity belongs to an external toxin or disease-injury model where the oligo is only a knockdown tool, or if "cytotoxicity" is a cancer-cell-killing efficacy readout. - *offtarget domain:* ACCEPT only an OBSERVED unintended-effect result: seed-mediated, mismatch/hybridization-dependent, transcriptome-wide, RNA-seq, or microarray off-target evidence. REJECT computational/design-only specificity screens, PK/biodistribution, and on-target/efficacy "specificity".

GATE 4 - GROUNDING. Provide grounding_quote: an EXACT verbatim span copied from the supplied passage that simultaneously establishes the in-scope oligo and the domain-matched primary result. If no single passage supports the accept, set grounding_quote to "NONE" and do not accept.

If the passage is too truncated/ambiguous to apply the gates, set decision="abstain".

GRADE (only when decision=accept): A = full-text Results/Figure/Table/Supplement section AND a NAMED therapeutic oligo AND correct evidence type; B = therapeutic oligo but abstract-only or weaker location; C = in-scope but generic/unnamed oligo or weak support.

S1.1 Grounding-gate enforcement (verbatim Python from `scripts/curate_v2_llm.py`)

```
def verify_grounding(decision: dict, passage: str) -> dict:
 """Force reject if an accept is not grounded in a verbatim substring of the passage."""
 if decision.get("decision") != "accept":
 return decision
 quote = (decision.get("grounding_quote") or "").strip()
 norm_p = " ".join(passage.split()).lower()
 norm_q = " ".join(quote.split()).lower()
 grounded = bool(norm_q) and norm_q != "none" and len(norm_q) >= 12 and norm_q in norm_p
 decision["grounding_verified"] = grounded
 if not grounded:
 decision["decision"] = "reject"
 decision["grade"] = "NA"
 decision["reason"] = "ungrounded: grounding_quote is not a verbatim span of the source passage
 return decision
```

The check is intentionally strict: whitespace is collapsed, comparison is case-insensitive, and a minimum

12-character quote is required. An LLM that confabulates a rationale cannot pass this gate, because no such verbatim span exists in the cached source.

S1.2 Stage 3 human-adjudication contract

The LLM **never** writes to `curation_audit.curator_id`, `curator_decision`, or `validation_status='curator_veri`
The downstream promoter (`scripts/promote_curator_review.py`) hard-requires `validation_status='curator_ve`
so no machine output can reach the release tables without a human accept. The sole human curator (Ni Jie, University of Innsbruck) adjudicated each LLM proposal row by row.

S2. Database schema (text tree)

The release schema is the single SQLite file `data/oligosafety.db` (full DDL at `data/schema_sqlite.sql`). The text-tree below shows the in-scope tables and the foreign-key paths a downstream user follows from a primary source to released evidence; `curation_audit` is shown on the side as it cross-references every other table by `(entity_table, entity_id)`.

```
source_document (36,245 rows; PMID / PMCID / DOI / title / venue / license)
 |
 v
curation_queue (70,283 rows; per-(source, evidence_domain) task)
 |
 v
curation_candidate (41,114 rows; matched_terms / source_location / status
 in { machine_precurated_v1,
 candidate_needs_curator_review,
 curator_rejected,
 recurated_rejected })
 |
 +--> toxicity_endpoint (release: 626 curator-verified rows)
 | molecule_id --> molecule
 | assay_id --> assay
 | source_document_id --> source_document
 | endpoint_name / endpoint_category / direction /
 | significance_label / is_observed_experimental /
 | source_location / evidence_grade
 |
 +--> offtarget_evidence (release: 111 curator-verified rows)
 molecule_id --> molecule
 assay_id --> assay
 source_document_id --> source_document
 offtarget_gene_symbol / offtarget_transcript_id /
 evidence_type / match_type / seed_match_length /
 is_observed_experimental / is_computational_prediction /
 source_location / evidence_grade

molecule (1,012 rows in the current molecule table)
```

```
 modality_id --> modality (e.g., ASO, siRNA, PMO, CpG ODN, aptamer)
 canonical_name / target_gene_symbol / disease_context /
 therapeutic_status / external_ids /
 sense_sequence / antisense_sequence / guide_sequence /
 passenger_sequence / seed_region /
 backbone_chemistry / sugar_modification / base_modification /
 conjugate_delivery /
 sequence_annotation_status / modification_annotation_status

assay (per-experiment context: species, route, dose, duration)
modality (controlled vocabulary; in_core_scope flag)

benchmark_split (344 rows; task_name {toxicity_safety_v0_1, offtarget_safety_v0_1};
 split_name {train, validation, test};
 entity_table + entity_id -> toxicity_endpoint / offtarget_evidence;
 split_strategy = 'source_plus_molecule_grouped_*';
 leakage_group keyed at (source_id, molecule_id) pair)

---- audit cross-cut ---------------------------------------------
curation_audit (one row per curator decision)
 entity_table { 'toxicity_endpoint' | 'offtarget_evidence' | 'curation_candidate' }
 entity_id (FK into the addressed table)
 validation_status
 { 'curator_verified' -- 737 current release rows
 , 'curator_rejected' -- 28,908 rows (mixed source)
 , 'machine_precurated_v1' -- 1,983 historical rows (never released)
 , 'recurated_rejected' -- 1,345 rows (demoted after source review)
 , 'verified' -- 3 rows, editorial_seed_nonhuman }
 curator_decision { 'accept' | 'reject' | NULL for machine rows }
 curator_id { human curator id | 'machine_v1_keyword_classifier' |
 'editorial_seed_nonhuman' | NULL }
 audit_note / audited_at
```

The candidate-to-release firewall is enforced by:

1. status enums (`machine_precurated_v1` is never `'curator_verified'`),
2. the `promote_curator_review.py` precondition (only `'curator_verified'` rows enter the release tables), and
3. an automated QA assertion in `scripts/final_delivery_check.py` that every release row has a matching `validation_status='curator_verified'` audit by a human curator id.

S3. Comparator matrix

Dimension	theRNA (NAR 2026)	siRNAEfficacyDB (IET 2024)	CMsiRNAdb (BMC 2026)	siRNAmod (Scientific Reports 2016)	CRISPRoffT (NAR 2025)	**OligoVigil (this work)**
Primary scope	Broad functional RNA therapeutics catalogue	siRNA on-target silencing efficacy	Chemically modified siRNA efficacy	Chemically modified siRNA catalogue (efficacy)	CRISPR/Cas guide off-targets (different molecular class)	**Safety + off-target evidence for therapeutic oligonu-cleotides (ASO / siRNA / PMO / LNA / aptamer / GalNAc-siRNA)**
Evidence type	Curated efficacy entries	Silencing efficacy measure-ments	Modification-aware efficacy	Modification annotations + efficacy	Off-target observations for Cas9/Cas12	**Observed safety endpoints + observed off-target results (737/737 observed experimen-tal rows)**
Source anchoring (exact in-source location)	Reference-level	Reference-level	Reference-level	Reference-level	Reference-level	**Source-localised: section / figure / table / paragraph captured per release row; 74.2% (547/737) full-text PMC-anchored; 100% PMID, 99.5% DOI**

Dimension	theRNA (NAR 2026)	siRNAEfficacyDB (IET 2024)	OMsiRNAdb (BMC 2026)	siRNAmod (Scientific Reports 2016)	CRISPRoffT (NAR 2025)	**OligoVigil (this work)**
Audit trail (machine-vs-human separation)	Not provided	Not provided	Not provided	Not provided	Not provided	**Three-stage pipeline: candidate -> v1 machine pre-curation -> v2 source-grounded LLM proposal -> single-human accept/reject + grade; full `curation_audit` table downloadable; 1,345 demoted candidates retained as `recurated_rejected`**
Benchmark splits	Not provided	Not provided	Not provided	Not provided	Not provided	**344 Grade A/B records split 218/23/22 (toxicity) + 66/5/10 (off-target); `source_plus_molecule_gr` strategy; 4 deterministic prior baselines provided**

Dimension	theRNA (NAR 2026)	siRNAEfficacyDB (IET 2024)	CMsiRNAdb (BMC 2026)	siRNAmod (Scientific Reports 2016)	CRISPRoffT (NAR 2025)	**OligoVigil (this work)**
No-login web access	Browser UI (catalogue)	Browser UI	Browser UI	Browser UI	Browser UI	**No-login portal + documented REST API + OpenAPI + MCP server +** `llms.txt` **+ Bioschemas JSON-LD + W3C PROV; bulk CSV/ZIP downloads**

Bottom line: none of the comparators ship a curator-verified, source-anchored, graded, audited safety/off-target evidence layer with a leakage-aware benchmark. OligoVigil targets a complementary niche to all five.

S4. Reproducibility - deterministic prior baselines on the fixed splits

Source: `data/generated/benchmark_baseline_results_v1.csv` (16 rows = 4 baselines x 2 tasks x 2 evaluation splits = validation + test). Reproduced verbatim below.

task_name	baseline_model	split	train n	eval n	majority_label	majority_frac_train	accuracy	macro-F1	coverage
offtarget_safety_v0	train_majority_class	validation	66	5	seed-mediated off-target effect	0.3788	0.2000	0.1111	1.00
offtarget_safety_v0	modality_prior_class	validation	66	5	seed-mediated off-target effect	0.3788	0.2000	0.1250	1.00

task_name	baseline_model	split	train n	eval n	majority_label	majority_frac_train	accuracy	macro-F1	coverage
offtarget_safety_v0_1	evidence_grade_prior_class	validation	66	5	seed-mediated off-target effect	0.3788	0.2000	0.1111	1.00
offtarget_safety_v0_1	target_prior_class	validation	66	5	seed-mediated off-target effect	0.3788	0.2000	0.1111	1.00
offtarget_safety_v0_1	train_majority_class	test	66	10	seed-mediated off-target effect	0.3788	0.4000	0.1429	1.00
offtarget_safety_v0_1	modality_prior_class	test	66	10	seed-mediated off-target effect	0.3788	0.6000	0.3667	0.90
offtarget_safety_v0_1	evidence_grade_prior_class	test	66	10	seed-mediated off-target effect	0.3788	0.4000	0.1429	1.00
offtarget_safety_v0_1	target_prior_class	test	66	10	seed-mediated off-target effect	0.3788	0.4000	0.1429	0.90
toxicity_safety_v0_1	train_majority_class	validation	218	23	hepatic	0.4954	0.6087	0.1261	1.00
toxicity_safety_v0_1	modality_prior_class	validation	218	23	hepatic	0.4954	0.5652	0.1204	1.00
toxicity_safety_v0_1	evidence_grade_prior_class	validation	218	23	hepatic	0.4954	0.6087	0.1261	1.00
toxicity_safety_v0_1	target_prior_class	validation	218	23	hepatic	0.4954	0.6087	0.1261	0.9565
toxicity_safety_v0_1	train_majority_class	test	218	22	hepatic	0.4954	0.6364	0.2593	1.00
toxicity_safety_v0_1	modality_prior_class	test	218	22	hepatic	0.4954	0.6364	0.2593	0.9545
toxicity_safety_v0_1	evidence_grade_prior_class	test	218	22	hepatic	0.4954	0.6364	0.2593	1.00
toxicity_safety_v0_1	target_prior_class	test	218	22	hepatic	0.4954	0.6364	0.2593	0.7727

S4.1 One-line summary per baseline

- `train_majority_class` - predicts the global training-set majority label for every evaluation row. Sanity floor. Coverage 1.00 by construction.
- `modality_prior_class` - predicts the training-set majority label *within the evaluation row's modality*; falls back to the global majority when the modality is unseen. Coverage drops below 1.00 only

when the held-out test row contains a modality with no training-set support (one off-target test row at coverage 0.90; one toxicity test row at coverage 0.9545).

- `evidence_grade_prior_class` - predicts the training-set majority label *within the same evidence grade* (A or B). Acts as a difficulty diagnostic: identical to `train_majority_class` here because the train-set majority label is the same in both grades.
- `target_prior_class` - predicts the training-set majority label *within the same target gene symbol*; coverage falls when the test row's target is unseen in training (off-target test 0.90; toxicity test 0.7727), serving as a leakage check.

All four baselines tie at macro-F1 = 0.2593 on the toxicity test set (n=22), reflecting the dominance of the "hepatic" class and the small n; off-target test macro-F1 spans 0.1429-0.3667 (n=10). These are reported only as difficulty diagnostics, not as method benchmarks.

S5. Provenance of `v2_human_override_decisions.csv`

`04_delivery/v2_human_override_decisions.csv` provides the per-candidate decision triple for every row in the v1 pre-curation pool, so the human-vs-model statistics in Methods Stage 3 are reproducible. Schema (14 columns, 2,003 rows):

column	description
`candidate_id`	stable id of the v1 candidate (<entity_table>:<entity_id>)
`entity_table`	`toxicity_endpoint` or `offtarget_evidence`
`entity_id`	row id in the addressed release table
`pmid`	source PubMed id
`doi`	source DOI (99.7% populated)
`source_location`	exact in-source pointer (section / paragraph / figure / table)
`v1_keyword_decision`	v1 keyword classifier verdict (`accept` / `reject`)
`v2_llm_proposal`	v2 source-grounded LLM verdict (`accept` / `reject` / `abstain`)
`human_decision`	Ni Jie's final verdict (`accept` / `reject`)
`human_grade`	final grade (A/B/C/NA) - only for accepts
`is_observed_experimental`	0/1, copied from the release table after promotion
`is_computational_prediction`	0/1 (only the single Grade-C off-target row id 156 is 1)
`current_validation_status`	`curator_verified` for accepts; `recurated_rejected` for human rejects of v1 accepts; `curator_rejected` for v1 rejects
`curator_id`	`ni_jie` for every human-touched row

S5.1 v2 LLM proposal x human decision cross-tabulation (n = 2,003)

	human=accept	human=reject	row total
v2=accept	618	**20**	638

	human=accept	human=reject	row total
v2=reject	**7**	523	530
v2=abstain	**33**	802	835
column total	658	1,345	2,003

Highlighted cells reproduced exactly:

- v2 accept x human reject = **20** (over-accepts caught)
- v2 reject x human accept = **7** (over-rejects recovered)
- v2 abstain x human accept = **33** (abstains promoted on reading)

Firm-decision overrides = 20 + 7 = 27 of 1,168 firm calls = **2.3%** divergence (Methods Stage 3). The 33 abstain-to-accept recoveries are reported separately because abstain is the model's explicit "ask a human" signal, not a substantive disagreement.

The historical Stage-3 decision table contains **658 human accepts and 1,345 rejects** from the original 2,003-candidate pool. One computational accept was subsequently removed, leaving 657 rows from that pool; later curator-verified expansion rounds added 80 observed rows, producing the current 737-row release.

S5.2 Cell-count re-verification command

```
import csv
from collections import Counter
with open("04_delivery/v2_human_override_decisions.csv", encoding="utf-8") as fh:
 rows = list(csv.DictReader(fh))
c = Counter()
for r in rows:
 c[(r["v2_llm_proposal"], r["human_decision"])] += 1
# Reproduces the table above exactly (2,003 rows, 6 occupied cells).
```

S6. Backup chain - DB integrity restoration points

Each backup is a `.bak` snapshot of `data/oligosafety.db` taken immediately before a curation-integrity edit, so any contested edit can be inspected by diff. Output of `ls data/oligosafety.db.pre_*.bak` (sorted lexicographically; 14 snapshots):

```
data/oligosafety.db.pre_audit_reconcile_20260607_052436.bak
data/oligosafety.db.pre_batch008_20260602.bak
data/oligosafety.db.pre_batch009_mega_fast_20260602.bak
data/oligosafety.db.pre_candidate_enum_20260607_025745.bak
data/oligosafety.db.pre_curator_identity_20260607_032934.bak
data/oligosafety.db.pre_enum_rename_20260606_122249.bak
data/oligosafety.db.pre_enum_rename_20260607_022138.bak
data/oligosafety.db.pre_full_rebuild_20260607_022009.bak
data/oligosafety.db.pre_garbage_molecule_fix_20260607_052332.bak
data/oligosafety.db.pre_status_relabel_20260604.bak
data/oligosafety.db.pre_leakage_relabel_20260607_053524.bak
```

```
data/oligosafety.db.pre_recuration_demote_20260606_121506.bak
data/oligosafety.db.pre_recuration_demote_20260606_121512.bak
data/oligosafety.db.pre_recuration_demote_20260607_022035.bak
```

S6.1 What each backup precedes

- `pre_batch008_20260602.bak`, `pre_batch009_mega_fast_20260602.bak` - pre-batch candidate-ingestion snapshots from the June-02 release-scale pre-curation runs.
- `pre_status_relabel_20260604.bak` - pre-relabel snapshot before the v1 status enum was renamed to make `machine_precurated_v1` distinct from any human verdict.
- `pre_enum_rename_20260606_122249.bak`, `pre_enum_rename_20260607_022138.bak` - pre-rename snapshots for the candidate-status enum normalisations.
- `pre_recuration_demote_20260606_121506.bak`, `pre_recuration_demote_20260606_121512.bak`, `pre_recuration_demote_20260607_022035.bak` - snapshots immediately before the v2+human-driven demotions wrote `validation_status='recurated_rejected'` on the 1,345 unsupported candidates.
- `pre_full_rebuild_20260607_022009.bak` - pre-rebuild snapshot before the 2026-06-07 historical 658-record provisional release.
- `pre_candidate_enum_20260607_025745.bak` - pre-cleanup snapshot before the final candidate-enum migration.
- `pre_curator_identity_20260607_032934.bak` - snapshot taken before the curator-id merge (legacy curator labels were collapsed to canonical `ni_jie`).
- `pre_garbage_molecule_fix_20260607_052332.bak` - snapshot taken before merging 45 v1-extraction-artefact molecule_ids into 4 placeholder `unspecified <modality>` molecules (S7 item 2).
- `pre_audit_reconcile_20260607_052436.bak` - snapshot before pruning duplicate audit rows and 2 orphan audits, leaving one curator-verified accept audit per then-current release row; the current release contains 737 such audits.
- `pre_leakage_relabel_20260607_053524.bak` - snapshot before re-keying `benchmark_split.leakage_group` to canonical placeholder identifiers and dropping the offending test row from the cross-split (source x molecule) group (Methods, Benchmark construction; S7 item 3).

S6.2 Source-license / reuse-category per-class counts

Query:

```sql
SELECT license_status, reuse_category, COUNT(*) FROM source_document
 GROUP BY license_status, reuse_category ORDER BY 3 DESC;
```

Table S6.2a - all 36,245 indexed sources:

license_status	reuse_category	n
abstract_metadata_only	derived_annotations_only	36,238
open_access	query_linkout_only	3
cc_by_nc_nd	query_linkout_only	1
official_guideline	derived_annotations_only	1
official_notice	derived_annotations_only	1

license_status	reuse_category	n
open_access	derived_annotations_only	1
total		**36,245**

Table S6.2b - 660 current release-anchored sources:

license_status	reuse_category	n
abstract_metadata_only	derived_annotations_only	660
total		**660**

Interpretation: the indexed pool is overwhelmingly PubMed-abstract-derived (36,238 / 36,245 = 99.98%) with derived-annotations-only redistribution scope. The 7 exceptions are open-access full-text or official-guideline / official-notice sources that we either link out to or extract under derived-annotations-only terms. **Every release-anchored source falls under `abstract_metadata_only / derived_annotations_only`**, so the entire release rests on the most conservative reuse scope: we do not redistribute raw third-party article text or full-text PDFs; we only store derived annotations (canonical name, endpoint label, evidence grade) and a verbatim grounding-quote span of the length permitted under fair-use / quotation right, together with the exact source location (section / paragraph / figure / table) so users can retrieve the original under their own subscription.

S7. Excluded and residual-record inventory

The current release separates removed rows, residual metadata gaps and benchmark exclusions so that users can reproduce each boundary.

S7.1 Removed computational off-target row

An earlier release-candidate table contained one Grade-C computational off-target prediction (`offtarget_evidence.id = 156`). It was removed from the release during curation review and was not re-introduced. Consequently, **all 737 current release rows are observed experimental results**.

S7.2 Residual placeholder molecules

The collaborator B2 recovery pass resolved 110 of 143 v5 placeholder-linked release rows. The current disclosed residual is **31 release rows** on true placeholder or mixed-modality fallback molecules. Within the frozen 344-row benchmark, **14 rows (4.1%)** remain attached to placeholders, down from 107/344 (31.1%) at v5. Users requiring named-molecule isolation should restrict analysis to the 330-row named-molecule benchmark subset.

S7.3 Grade A/B release rows outside the frozen benchmark

The release contains **508 Grade A/B records**, of which **344** are assigned to the frozen reference benchmark. The remaining **164 Grade A/B rows** are released as evidence but are not benchmark-assigned because they are singleton leakage groups or await promotion of later expansion batches under the pair-level isolation rule.

S7.4 Pair-level leakage invariant

The current benchmark enforces pair-level isolation at `(source_document_id, molecule_id)`. The release check:

```sql
SELECT leakage_group, COUNT(DISTINCT split_name)
FROM benchmark_split
GROUP BY leakage_group
HAVING COUNT(DISTINCT split_name) > 1;
```

returns no rows. The benchmark is therefore pair-isolated, while the manuscript explicitly discloses that strict molecule-level isolation requires the 330-row named-molecule subset.

S8. Closest-work feature audit underlying Figure 5

Figure 5 separates literature overlap from resource capability. Source-PMID sets were available for OligoVigil, CRISPRoffT and siRNAEfficacyDB; all pairwise and triple intersections were zero. Feature support was assessed conservatively as **yes**, **partial**, or **absent/undocumented** from the corresponding paper or inspectable portal.

Feature	OligoVigil	theRNA	siRNAEfficacyDB	OMsiRNAdb	siRNAmod	CRISPRoffT
Total curated records	yes	yes	yes	yes	yes	yes
Exact source location	yes	yes	partial	partial	partial	partial
Curator audit trail	yes	partial	absent	absent	absent	partial
Inter-curator check	yes	absent	absent	absent	absent	absent
Machine-stage FAR audit	yes	absent	absent	absent	absent	absent
Benchmark splits	yes	absent	partial	absent	absent	partial
Deterministic baselines	yes	absent	absent	absent	absent	absent
Structured assay metadata	partial	partial	yes	partial	absent	partial
Per-position chemistry	absent	partial	absent	yes	yes	absent
Off-target gene resolution	partial	absent	absent	absent	absent	yes
No-login portal	yes	yes	yes	yes	absent	yes
API / OpenAPI	yes	partial	partial	partial	absent	partial
Agent-readable metadata	yes	absent	absent	absent	absent	partial
Bulk download	yes	partial	partial	partial	partial	partial
Versioned release	yes	partial	partial	partial	partial	partial
Named maintainer	yes	yes	yes	yes	partial	yes

Interpretation. OligoVigil does not lead on absolute scale, per-position chemistry, dose coverage or genelevel off-target resolution. Its distinguishing combination is exact source-localized evidence, a downloadable human audit trail, measured machine-stage error, inter-curator reliability reporting, a frozen reference split and programmatic reuse surfaces.

The source-data tables used to render Figure 5 are:

- `figures/source_data/FIG5_closest_work_audit_v2_sets.csv`
- `figures/source_data/FIG5_closest_work_audit_v2_intersections.csv`
- `figures/source_data/FIG5_closest_work_audit_v2_features.csv`

End of Supplementary Materials.